1 Formal Theory: Why Compositional Factorization Separates Unseen Mixtures

This section explains *why* the library-novelty score S_{lib} of Section **??** detects unseen compositions, and *what exactly is and is not guaranteed*. Three kinds of statements are kept strictly apart: (a) a finite-sample, distribution-free guarantee on the in-distribution (ID) false-rejection rate (Lemma 1); (b) a *deterministic, conditional* separation bound that converts primitive-detector margins into a lower bound on S_{lib} (Theorem 1, Corollary 1, Proposition 2); and (c) a structural statement about what a flat softmax can represent (Proposition 3). Statement (b) is *not* a probabilistic guarantee of out-of-distribution (OOD) detection: it holds only when the primitive head satisfies the stated margin condition on the wafer in question.

1.1 Setup and notation

Let $\mathcal{P} = \{\text{C}, \text{D}, \text{EL}, \text{ER}, \text{L}, \text{NF}, \text{S}, \text{RAND}\}$ be the 8 defect primitives, indexed $k = 0, \dots, 7$, and let $z \in \{0,1\}^8$ be an attribute vector. The physical grammar admits the set $\mathcal{V} \subset \{0,1\}^8$ of configurations satisfying $z_{\text{C}} z_{\text{D}} = 0$, $z_{\text{EL}} z_{\text{ER}} = 0$ and $(z_{\text{NF}} \vee z_{\text{RAND}}) \Rightarrow \sum_k z_k = 1$; one checks by enumeration that $|\mathcal{V}| = 38$. The *known vocabulary* is $\mathcal{V}_{\text{known}} = \{z \in \mathcal{V} : \ \|z\|_1 \le c\}$ with $c = 3$, so $|\mathcal{V}_{\text{known}}| = 34$ (classes C1–C34), while the four quadruple-mixtures C35–C38 are exactly the elements of $\mathcal{V}$ with $\|z\|_1 = 4$ and lie outside $\mathcal{V}_{\text{known}}$.

The primitive head outputs logits $l(x) \in \mathbb{R}^8$ and induces the factorized distribution

$$q_\theta(z \mid x) = \prod_{k=0}^{7} \sigma(l_k(x))^{z_k} \big(1 - \sigma(l_k(x))\big)^{1 - z_k}, \qquad z \in \{0,1\}^8, \tag{1}$$

which is a probability distribution on all 2^8 vectors (not only on $\mathcal{V}$). The *library-novelty score* and the induced *library set* are

$$S_{\text{lib}}(x) = -\log \sum_{z \in \mathcal{V}_{\text{known}}} q_\theta(z \mid x), \qquad C_{\text{lib}}(x) = \begin{cases} \varnothing & S_{\text{lib}}(x) > \hat{q}_{\text{lib}}, \\ \text{(in-library)} & \text{otherwise}, \end{cases} \tag{2}$$

where "$C_{\text{lib}}(x) = \varnothing$" is shorthand for "$x$ is flagged as out-of-library" and $\hat{q}_{\text{lib}}$ is calibrated as follows. Given N ID calibration scores $S_1, \dots, S_N$ (S_{lib} evaluated on held-out ID wafers) and a level $\alpha \in (0,1)$, let $\hat{q}_{\text{lib}} = S_{(k)}$ be the k-th smallest calibration score, $k = \lceil (N+1)(1-\alpha) \rceil$. Here $N = 6{,}803$ and $\alpha = 0.05$, so $k = 6{,}464$.

Lemma 1 (ID false-rejection control)**.** *Assume the network parameters θ and the vocabulary $\mathcal{V}_{\text{known}}$ are fixed independently of the calibration and test wafers, and that a new ID wafer X_{N+1} is exchangeable with the calibration wafers. Then $\mathbb{P}\big(S_{\text{lib}}(X_{N+1}) > \hat{q}_{\text{lib}}\big) \le \alpha$.*

Proof. By exchangeability (and, with ties, a standard randomization/worst-case argument), the rank of S_{N+1} among $S_1, \dots, S_{N+1}$ is uniform on $\{1, \dots, N+1\}$ up to ties that only decrease the rejection probability. The event $S_{N+1} > S_{(k)}$ requires rank $> k$, which has probability at most $(N+1-k)/(N+1) \le \alpha$. □

This is the conformal-outlier construction of Bates et al. [**?**] (see also [**?**, **?**]). It controls the *marginal* (averaged over calibration draws) ID false-rejection rate only; the realized rate on a finite test set fluctuates around α (we report Wilson intervals) and no statement about OOD power is implied. Power is the subject of the remainder of this section.

1.2 A margin condition and the separation bound

For a wafer x whose true composition is $z^\star \in \mathcal{V}$, let $A = \{k : z^\star_k = 1\}$ be its active primitives and $n = |A|$.

Definition 1 (τ-margin on the active set)**.** The primitive head has *margin* $\tau \in \mathbb{R}$ *on* x if $l_k(x) \geq \tau$ for every $k \in A$. Write $\varepsilon_\tau = \sigma(-\tau) = 1/(1+e^\tau) \leq e^{-\tau}$.

Note that τ may be negative: $\tau = 0$ means every active primitive has detector probability at least $1/2$. The condition concerns the network's *output* on x; it is a property of the trained model, not of the label.

Theorem 1 (Compositional separation bound)**.** *Let x have active set A with $n = |A|$ and $M = n - c \geq 1$, and suppose the head has margin τ on x. Then*

$$\textit{(i)} \quad \sum_{z \in \mathcal{V}_{\text{known}}} q_\theta(z \mid x) \;\leq\; \binom{n}{M}\varepsilon_\tau^M \;\leq\; \binom{n}{M} e^{-M\tau}, \tag{3}$$

$$\textit{(ii)} \quad S_{\text{lib}}(x) \;\geq\; M\log\bigl(1+e^\tau\bigr) - \log\binom{n}{M} \;\geq\; M\tau - \log\binom{n}{M}, \tag{4}$$

$$\textit{(iii)} \quad S_{\text{lib}}(x) \;\geq\; -\log\bigl(1-\sigma(\tau)^n\bigr) \;\geq\; \sigma(\tau)^n. \tag{5}$$

(In the present ontology $n \leq 4$, so $M = 1$ is the only case that occurs; the proof of (iii) holds for every $M \geq 1$.)

Proof. Let $Z \sim q_\theta(\cdot \mid x)$; its coordinates are independent Bernoulli variables with $\mathbb{P}(Z_k = 1) = \sigma(l_k)$, and $\sum_{z \in \mathcal{V}_{\text{known}}} q_\theta(z \mid x) = \mathbb{P}(Z \in \mathcal{V}_{\text{known}})$.

(i) Every $z \in \mathcal{V}_{\text{known}}$ has $\|z\|_1 \leq c$, hence $|z \cap A| \leq c$ and $|A \setminus z| \geq n - c = M$. Therefore $\{Z \in \mathcal{V}_{\text{known}}\} \subseteq \bigcup_{S \subseteq A,\, |S| = M} \{Z_k = 0 \;\forall k \in S\}$. A union bound over the $\binom{n}{M}$ subsets and independence give

$$\mathbb{P}(Z \in \mathcal{V}_{\text{known}}) \leq \sum_{|S|=M} \prod_{k \in S} \bigl(1 - \sigma(l_k)\bigr) = \sum_{|S|=M} \prod_{k \in S} \sigma(-l_k) \leq \binom{n}{M}\varepsilon_\tau^M,$$

because $\sigma(-\cdot)$ is decreasing and $l_k \geq \tau$ on A. Finally $1/(1+e^\tau) \leq e^{-\tau}$.

(ii) Apply $-\log$ (decreasing) to (i): $-\log\bigl(\binom{n}{M}\varepsilon_\tau^M\bigr) = M\log(1+e^\tau) - \log\binom{n}{M}$, and $\log(1+e^\tau) \geq \tau$.

(iii) If $Z \supseteq A$ then $\|Z\|_1 \geq n > c$, so $Z \notin \mathcal{V}_{\text{known}}$. Hence $\mathbb{P}(Z \in \mathcal{V}_{\text{known}}) \leq 1 - \mathbb{P}(Z \supseteq A) = 1 - \prod_{k \in A} \sigma(l_k) \leq 1 - \sigma(\tau)^n$, since σ is increasing. Applying $-\log$ gives the first inequality, and $-\log(1-u) \geq u$ gives the second. □

Remark 1 (Two regimes). Bounds (i)–(ii) come from a union bound and are informative only for large margins: (ii) is vacuous whenever $M\log(1+e^\tau) \leq \log\binom{n}{M}$, which for $n = 4$, $M = 1$ means $\tau \leq \log 3 \approx 1.10$. They show the qualitative law: the library mass decays *exponentially* and S_{lib} grows *linearly* in the margin. Bound (iii) uses the joint event that all n primitives fire, is non-vacuous for any τ, and is the one used for threshold statements.

Corollary 1 (Margin sufficient for rejection)**.** Let $n = 4$, $M = 1$ and let $\hat{q}_{\text{lib}} > 0$ be the calibrated threshold. Define

$$\tau^*(\hat{q}_{\text{lib}}) = \sigma^{-1}\Bigl(\bigl(1 - e^{-\hat{q}_{\text{lib}}}\bigr)^{1/4}\Bigr).$$

If the head has margin $\tau > \tau^*(\hat{q}_{\text{lib}})$ on x, then $S_{\text{lib}}(x) > \hat{q}_{\text{lib}}$, i.e. x is rejected as out-of-library. For $\hat{q}_{\text{lib}} = 0.0101$, $\tau^* \approx -0.769$.

Proof. By (iii), $S_{\text{lib}}(x) \geq -\log(1-\sigma(\tau)^4)$, which is strictly increasing in τ and equals $\hat{q}_{\text{lib}}$ at $\tau = \tau^*$ (solve $1-\sigma(\tau)^4 = e^{-\hat{q}_{\text{lib}}}$). □

Reading of the corollary. A quad-mixture is rejected whenever *all four* of its constituent primitives are detected with probability above $\sigma(\tau^*) \approx 0.316$. The detection requirement is deliberately weak, which is why the factorized score is effective; conversely the corollary identifies the failure mode precisely: a quad-mixture can escape only if at least one constituent primitive falls below τ^*. This matches the observation that the 20 unrejected wafers of C35–C38 all have a Loc or Scratch logit below $-0.77 < \tau^*$ (Section **??**). The numerical value of τ^* must be computed from the *unrounded* $\hat{q}_{\text{lib}}$; because τ^* increases with $\hat{q}_{\text{lib}}$, a rounded -0.77 lies slightly *outside* the sufficient region.

1.3 Known patterns are not penalized

Proposition 2 (Low score on in-library patterns)**.** *Let $z^\star \in \mathcal{V}_{\text{known}}$ with $A = \{k : z^\star_k = 1\}$, $|A| = n$, and suppose $l_k \geq \tau$ for $k \in A$ and $l_k \leq -\tau'$ for $k \notin A$. Then*

$$0 \leq S_{\text{lib}}(x) \;\leq\; -\log q_\theta(z^\star \mid x) \;\leq\; n\,\text{softplus}(-\tau) + (8-n)\,\text{softplus}(-\tau') \;\leq\; ne^{-\tau} + (8-n)e^{-\tau'},$$

which tends to 0 as $\tau, \tau' \to \infty$.

Proof. Since $z^\star \in \mathcal{V}_{\text{known}}$, $\sum_{z \in \mathcal{V}_{\text{known}}} q_\theta \geq q_\theta(z^\star \mid x)$, so $S_{\text{lib}} \leq -\log q_\theta(z^\star \mid x)$. Because $-\log\sigma(l) = \text{softplus}(-l)$ and $-\log(1-\sigma(l)) = \text{softplus}(l)$, $-\log q_\theta(z^\star \mid x) = \sum_{k\in A}\text{softplus}(-l_k) + \sum_{k\notin A}\text{softplus}(l_k)$, and softplus is increasing with $\text{softplus}(t) \leq e^t$. Non-negativity holds as $\sum_z q_\theta \leq 1$. □

Proposition 2 says that well-detected library patterns are not rejected *by the score*; it does not by itself bound the ID false-rejection rate, which is controlled by calibration (Lemma 1) and is reported empirically in Section **??**.

1.4 Why a flat softmax cannot express library novelty

Proposition 3 (Closed-world normalization)**.** *Let a flat classifier output $p(\cdot \mid x) \in \Delta^{33}$ over the 34 known classes. Then (a) its predictive mass on the library is $p(\mathcal{V}_{\text{known}} \mid x) = \sum_y p(y \mid x) = 1$ for every x, so the event "$Y \notin \mathcal{V}_{\text{known}}$" has probability zero under the model; and (b) any scoring rule built from the flat head is a function $s = g(p(\cdot \mid x))$ of the simplex point alone.*

Proof. (a) is the normalization constraint. (b) holds by definition of a score computed from the head output. □

Consequences. Under the flat head the quantity that S_{lib} measures is identically 0; any OOD score derived from it (e.g. $1 - \max_y p$, as in the conformal baseline) must therefore exploit the *shape* of p rather than a probability of novelty, and it cannot distinguish a quad-mixture whose mass splits across two known neighbours from an ID wafer that is genuinely ambiguous between those same neighbours, whenever their softmax outputs coincide. Proposition 3 does *not* claim that softmax scores carry no OOD signal (the baseline attains 65.13% rejection); it states that the signal is indirect and bounded by what the simplex retains. By contrast, (1) places mass on $\{0,1\}^8 \supsetneq \mathcal{V}_{\text{known}}$, so novelty is a measurable event with probability $1 - \sum_{z\in\mathcal{V}_{\text{known}}} q_\theta(z \mid x)$.

1.5 Scope, relation to the discrete rule, and attribution

Remark 2 (Grammar is not needed for the bound). Theorem 1 uses only the cardinality cap c; the mutual-exclusion and solitary constraints enter only through the definition of $\mathcal{V}_{\text{known}}$. Removing configurations from $\mathcal{V}_{\text{known}}$ can only decrease $\sum_{z\in\mathcal{V}_{\text{known}}} q_\theta$ and thus increase S_{lib}, so the bound is valid for any $\mathcal{V}_{\text{known}} \subseteq \{z : \|z\|_1 \le c\}$.

Remark 3 (Relation to the discrete symbolic rule). Let $R_{\text{rule}} = \{x : \sum_k \mathbb{I}(l_k(x) > 0) > 3\}$. If $x \in R_{\text{rule}}$, pick four indices with $l_k > 0$; the probability that all four fire is at least 2^{-4}, and any such Z has $\|Z\|_1 \ge 4$, so $S_{\text{lib}}(x) \ge -\log(1 - 2^{-4}) \approx 0.0645 > \hat{q}_{\text{lib}} = 0.0101$. Hence $R_{\text{rule}} \subseteq \{S_{\text{lib}} > \hat{q}_{\text{lib}}\}$: the calibrated score rejects *at least* everything the rule rejects, plus wafers with weaker evidence. This is consistent with the ablation (OOD: $770 \le 780$; ID: $279 \le 325$) and shows that the gain of S_{lib} over the rule is a *refinement* (extra 10 OOD detections at the cost of extra 46 ID rejections), whereas the gain of either over the flat head is due to the factorized primitive representation.

Remark 4 (Beyond the cardinality cap (not evaluated here)). If $z^\star \notin \mathcal{V}_{\text{known}}$ has Hamming distance $M = \min_{z\in\mathcal{V}_{\text{known}}} \|z - z^\star\|_0 \ge 1$ from the vocabulary and the head satisfies $|l_k| \ge \tau$ with the correct sign on *all* 8 coordinates, then $\{Z \in \mathcal{V}_{\text{known}}\} \subseteq \{\|Z - z^\star\|_0 \ge M\}$ and the same union bound gives $\sum_{z\in\mathcal{V}_{\text{known}}} q_\theta \le \binom{8}{M}\varepsilon_\tau^M$. This would cover unseen configurations of any depth provided $\mathcal{V}_{\text{known}}$ is defined as the set of configurations observed in training; testing it requires leave-one-configuration-out experiments that we do not report.

Limitations of the theory. Theorem 1 is conditional on the head's margins and is therefore a statement about *primitive recall*, not an unconditional OOD guarantee; its proof is an elementary union bound and we present it as an explanation of observed behaviour rather than as a deep result. The vocabulary cutoff $c = 3$ coincides with the unseen set in our benchmark, so the empirical evaluation on C35–C38 is an exploratory, post-hoc development result (Section **??**); novelty from a *new primitive* is outside the scope of the bound.

Attribution. The quantity $-\log \sum_{z\models\alpha} \prod_k p_k^{z_k}(1-p_k)^{1-z_k}$ is the *semantic loss* of Xu et al. [**?**] with the constraint α taken to be membership in $\mathcal{V}_{\text{known}}$; we use it at test time as a novelty score and calibrate it conformally rather than as a training regularizer. Predicting independent attributes and composing them into class posteriors follows Direct Attribute Prediction of Lampert et al. [**?**]. The threshold construction follows conformal outlier detection [**?**]. Our contributions are the wafer-specific grammar, the polar geometry, the margin analysis above, and the empirical evaluation—not the score functional itself.

2 Appendix A: Score–Probability Monotonicity and Reconciliation of the Rejection Counts

Purpose. Two different thresholds appear in Section **??**: a quantile $\hat{q}$ of a *nonconformity score* $s \in [0, 1]$ (large = anomalous), and a cutoff τ_p on the *maximum softmax probability* (small = anomalous). They are related by $\tau_p = 1 - \hat{q}$, so a *larger* score quantile corresponds to a *smaller* probability cutoff and hence to *fewer* rejections. Reading $\hat{q}$ as if it were a cutoff on $p_{\max}$ reverses this ordering and produces an apparent contradiction; this appendix removes it.

2.1 Definitions

Let $p(\cdot \mid x) \in \Delta^{K-1}$ be the softmax output of the frozen $K = 34$-class model (checkpoint `dapnet_34class_best.pt`) and $p_{\max}(x) = \max_y p(y \mid x)$. On the calibration set $\{(x_i, y_i)\}_{i=1}^{N}$, $N = 6{,}803$, define two score families:

$$s_{\text{conf}}(x_i, y_i) = 1 - p(y_i \mid x_i) \qquad \text{(true-label nonconformity; uses labels)}, \tag{6}$$

$$s_{\text{MSP}}(x_i) = 1 - p_{\max}(x_i) \qquad \text{(label-free)}. \tag{7}$$

With $k = \lceil (N+1)(1-\alpha) \rceil = 6{,}464$ for $\alpha = 0.05$, let $\hat{q}_{\text{conf}}$ and $\hat{q}_{\text{MSP}}$ be the k-th smallest value of each family. The conformal prediction set is $C(x) = \{y : \ 1 - p(y \mid x) \le \hat{q}_{\text{conf}}\}$ and the MSP rule rejects when $1 - p_{\max}(x) > \hat{q}_{\text{MSP}}$.

2.2 Three elementary facts

Lemma 2 (Empty set $\iff$ probability cutoff)**.** *For any $\hat{q} \in [0,1]$, $\{y : 1 - p(y \mid x) \le \hat{q}\} = \varnothing \iff p_{\max}(x) < 1 - \hat{q}$.*

Proof. The set is empty iff $1 - p(y \mid x) > \hat{q}$ for all y, i.e. $\min_y \big(1 - p(y \mid x)\big) = 1 - p_{\max}(x) > \hat{q}$, i.e. $p_{\max}(x) < 1 - \hat{q}$. □

Lemma 3 (Pointwise dominance and quantile order)**.** *For every i, $s_{\text{conf}}(x_i, y_i) \ge s_{\text{MSP}}(x_i)$, with equality iff y_i is an arg-max of $p(\cdot \mid x_i)$. Consequently $\hat{q}_{\text{conf}} \ge \hat{q}_{\text{MSP}}$.*

Proof. $p(y_i \mid x_i) \le p_{\max}(x_i)$ gives the pointwise inequality. If $a_i \ge b_i$ for all i, the k-th order statistic of (a_i) is at least that of (b_i), because for every level t the number of a_i below t is at most the number of b_i below t, so the k-th smallest a cannot be smaller than the k-th smallest b. □

Lemma 4 (Larger quantile ⇒ fewer rejections)**.** *Let $\tau_p = 1 - \hat{q}$ and $R(\tau_p) = \{x : p_{\max}(x) < \tau_p\}$. Then $\hat{q} \mapsto R(1 - \hat{q})$ is* non-increasing *in $\hat{q}$: $\hat{q}_1 \le \hat{q}_2 \Rightarrow R(1 - \hat{q}_2) \subseteq R(1 - \hat{q}_1)$.*

Proof. $\hat{q}_1 \le \hat{q}_2 \Rightarrow 1 - \hat{q}_2 \le 1 - \hat{q}_1$ and $\{p_{\max} < t\}$ grows with t. □

2.3 Reconciliation

Lemma 3 gives $\hat{q}_{\text{conf}} = 0.038288 \ \ge \ \hat{q}_{\text{MSP}} = 0.029358$ (the opposite ordering would be impossible). Lemma 4 then gives $R(0.961712) \subseteq R(0.970642)$: the conformal rule, whose quantile is larger, must reject *fewer* wafers than the MSP rule, on ID and on OOD alike. Table 1 shows the counts obey exactly this ordering. At *matched* probability cutoffs the two rules are the same decision rule (Lemma 2); they differ only because their calibration quantiles differ.

Table 1: Reconciled quantiles, cutoffs and rejection counts (frozen 34-class checkpoint; $N_{\text{cal}} = 6{,}803$, $N_{\text{ID}} = 6{,}803$, $N_{\text{OOD}} = 800$; 95% Wilson intervals).

Rule	$\hat{q}$	cutoff $\tau_p = 1 - \hat{q}$	ID rejected	OOD (C35–C38) rejected
Conformal (s_{conf})	0.038288	0.961712	294/6,803 = 4.32% [3.86, 4.83]	521/800 = 65.13% [61.76, 68.36]
MSP (s_{MSP})	0.029358	0.970642	343/6,803 = 5.04% [4.54, 5.59]	552/800 = 69.00% [65.73, 72.11]

2.4 Executable check

The following assertions, run on the saved calibration/test probability arrays, certify every claim above (they depend on no other result and fail loudly on any inversion).

```
import numpy as np, math
def kth(a, alpha):              # k-th smallest, k = ceil((N+1)(1-alpha))
    return np.sort(a)[math.ceil((len(a)+1)*(1-alpha)) - 1]
pc, yc = P_cal,  y_cal         # (N,34) probabilities, labels
pmax_c = pc.max(1)
q_conf = kth(1 - pc[np.arange(len(yc)), yc], 0.05)
q_msp  = kth(1 - pmax_c, 0.05)
assert q_conf >= q_msp                                    # Lemma 2
for q, P, name in [(q_conf, None, "conf"), (q_msp, None, "msp")]:
    tau = 1 - q
    rej_id  = (P_id.max(1)  <  tau)                       # Lemma 1: empty set <=> pmax < 1-q
    rej_ood = (P_ood.max(1) <  tau)
    if name == "conf":                                    # set-based definition must agree
        C_id  = (1 - P_id  <= q).sum(1) == 0
        C_ood = (1 - P_ood <= q).sum(1) == 0
        assert (C_id == rej_id).all() and (C_ood == rej_ood).all()
    print(name, round(q,6), round(tau,6), rej_id.sum(), rej_ood.sum())
# expected: conf 0.038288 0.961712 294 521 ; msp 0.029358 0.970642 343 552
assert 294 < 343 and 521 < 552                            # Lemma 3
```

Numerical sanity. $1-0.038288 = 0.961712$ and $1-0.029358 = 0.970642$, so $0.961712 < 0.970642$, matching $521 < 552$ and $294 < 343$.